

Allegheny County Tax · 3-Minute Demo Runbook

Snowflake Summit · Pediment Booth · for the recorded video walkthrough of the Allegheny County Tax dbt Wizard demo. Looped unattended on the booth screen.

3-5 days MANUAL MODEL BUILD TODAY	90 sec DBT WIZARD BUILD	\$2.8M ASSESSMENT REFUND EXPOSURE	18 hr EARLIER THAN APPEALS HEARING
--	--------------------------------------	--	---

ODI architecture · sources, lakehouse, consumers

Fivetran CDC connectors + event streams (Kafka / Kinesis for HL7, telematics, market data)	Iceberg + Glue Open lakehouse on S3, governed catalog, ACID snapshots, time travel	dbt Labs Bronze → Silver → Gold marts, committed as Iceberg snapshots	dbt Wizard Build-time AI · 90s authoring of governed gold marts	Snowflake + 4 Snowflake, Athena, Trino, DuckDB, Spark — same Parquet, no copies
--	--	--	--	--

One set of files on S3. Five engines read the same Parquet through the Glue catalog. Governed end-to-end: SOC 2 · CJIS · row-level access · column masking.

DATA ENGINEERS ANCHOR HERE Architecture · Wizard · Live · Pipeline Show the loop, the four sub-agents, the materialization, the connector deep-links. This is your DE proof.	BUSINESS USERS ANCHOR HERE Assessors, Appeals, Finance, Public Portal Show the answers, the KPI tiles, the segments, the executive read. The same governed gold layer powers all of it.
---	--

Frame the demo on business outcomes — never functionality

PAIN Chief Assessor can't get answers in time for the Appeals Board Hearing	REASON Every new question = 3-5 day DE backlog	IMPACT Missed committee, sized exposure, regulator scrutiny	CAPABILITY Build-time AI on the governed gold layer	PROOF Live model build in 90 seconds — on screen
---	--	--	---	--

The mantra: the win is not that dbt Wizard *can* author a model. The win is that the Chief Assessor walks into the next Appeals Board Hearing with a tested, lineage-tracked answer that did not exist this morning. Every beat below ties to that one sentence.

The 3-minute script • five beats

00:00 – 00:25 Beat 1 • The stakeholder hook

WHAT YOU SAY (25S)

"It is Friday morning at the County Assessor's office. The Chief Assessor needs to know: Why did 38% of appealed assessments in the new Eastside reval get reduced versus only 12% in the rest of the county? The Appeals Board Hearing is in 18 hours. Today that costs three-to-five days of DE work and a missed meeting."

WHAT YOU SHOW

Open the **Scenario page** (/scenario). The Chief Assessor's question is the first thing in frame. Let the T-minus counter tick.

OUTCOME → Anchors the audience on a real business question, not a feature.

00:25 – 00:50 Beat 2 • The cost of the gap

WHAT YOU SAY (25S)

"There is no gold-layer mart that answers this. Three days from now the team might land it. The Appeals Hearing will have already happened. A \$2.8 million assessment-refund exposure across the affected parcels — and that is just the dollars we can see."

WHAT YOU SHOW

Stay on Scenario. Highlight the four KPI tiles: 3-5 days, 90s, \$2.8M, T-minus to Appeals Board Hearing.

OUTCOME → Quantifies the cost of the gap. Sets up the 90-second proof.

00:50 – 02:20 Beat 3 • The live build (proof moment • DE-anchor)

WHAT YOU SAY (90S)

"We click Run the Build. Four sub-agents — Explorer, Summary, Worker, Verification — open the dbt project, find the right grain, pull a slice of the gold layer, write the new model, write its tests, and materialize it as an Iceberg snapshot. The viewer sees the SQL appear character by character. By the time we are done talking, the model exists, the tests pass, the lineage graph has updated."

Pause once at: "Worker is anchoring the CTE to the grain the Chief Assessor asked for — not a notebook-grade approximation."

And again at: "Verification is writing a uniqueness test on the natural key. This is a governed gold asset, not a SQL snippet."

WHAT YOU SHOW

Click into /wizard-live . Let the playback run at 1x. The agents type SQL and YAML in real time. The step rail moves Discovery → Schema → Inspection → Model Creation → Tests → Materialization. Terminal-window aesthetic.

OUTCOME → Data engineers see the loop end-to-end. The wow is that the output is a tested, governed asset — not a snippet pasted into Slack.

02:20 – 02:45 Beat 4 • The outcome (P&L tie-back • business-anchor)

WHAT YOU SAY (25S)

"The new mart is materialized. The Appeals Board Hearing convenes in 18 hr, not three days. The Chief Assessor walks in with the answer: 412 parcels in three Eastside neighborhoods share an outdated land-use code that overstates effective frontage; 17 of 22 successful appeals share that single coding signature. That is a code-table refresh, not a county-wide reval. And the next time anyone asks this — residential, industrial, mixed-use — the model is already there."

WHAT YOU SHOW

Cut to /outcome . Materialized model, tests passed, lineage updated, root-cause panel, impact panel. Then quick scroll through the business-user pages (Assessors, Appeals, Finance, Public Portal) so the audience sees the same gold layer powering the executive read.

OUTCOME → Closes the loop from question to action AND shows the business-user surface running on the same data.

02:45 – 03:00 Beat 5 • The architectural ask

WHAT YOU SAY (15S)

"Fivetran lands every source. Iceberg on S3 holds the bytes; the Glue catalog governs the snapshots. dbt Labs builds the marts. dbt Wizard authors the gaps. Snowflake reads the gold layer — and so do Athena, Trino, DuckDB, and Spark. One set of files, five engines. That loop closes for appeals attribution, reval transparency, public-portal reporting, and CJIS data-handling evidence — all on the same governed lake. Want to see how this maps to your gold layer? Stop by the Pediment booth or scan the QR."

WHAT YOU SHOW

Final card with the architecture loop, QR code to fivetran.com/contact. Tag dbt Labs co-brand.

OUTCOME → Calls out the architectural pattern. Invites a discovery conversation.

Business outcomes to weave through — never lead with features

OUTCOME THE BUYER FEELS	WHAT IT REPLACES	HOW TO PHRASE IT
Appeals Board Hearing gets the answer same-day	3-5 day DE backlog	In time for the meeting, not three days late
\$2.8M exposure sized in 90 seconds	Open gap window	The cost of the gap closed before lunch
Reusable governed mart for every future ask	One-off notebook scripts	The next person who asks gets it for free
CDO does not add headcount to keep up	Hiring DE backfill	Your team stops being the bottleneck
Auditor sees lineage + tests, not a query history	Manual evidence packs	Examiner-ready by default

Anti-patterns • what NOT to say

Don't say: "It runs dbt show against the warehouse and authors a new model file." **Say:** "Ninety seconds from question to materialized answer."

Don't say: "Cortex Analyst." Cortex is intentionally absent from this demo. The story is build-time AI on the governed lake.

Don't say: "dbt Wizard generates SQL." Say: "dbt Wizard authors a governed asset — schema contract, tests, lineage, ownership."

Don't say: "Look at the typing animation." The animation is the texture, not the value.

Don't open with the architecture diagram. Open with the Chief Assessor's question. Architecture is the closer.

Don't skip the business pages. After the wizard finishes, scroll through Assessors, Appeals, Finance, Public Portal so the audience sees the persona surface that runs on the same gold layer.

Discovery prompts to anchor before the demo

BEFORE YOU PRESS PLAY

- "What questions does your Chief Assessor ask that your team cannot answer same-day?"
- "When a stakeholder asks for a new cut of data, what is the average time from ask to merged dbt model?"
- "How many one-off SQL scripts are sitting in someone's Snowflake history that never made it back into the governed layer?"
- "Who is your Economic Buyer for new data investment? Likely audience: Chief Assessor, County CIO, Finance Director, Open-Government Officer."

AFTER THE DEMO LANDS

- "Which of those agent steps — Discovery, Schema, Inspection, Model Creation, Tests, Materialization — is where your team spends the most time today?"
- "If we sized a pilot on three of your highest-value recurring Chief Assessor questions, what would success look like in 90 days?"
- "Who needs to be in a 30-minute architecture review to see how this lands on your gold layer? Audience: Chief Assessor, County CIO, Finance Director, Open-Government Officer."

Personas at the booth — split the time

DATA ENGINEER · 60% OF BOOTH TIME

Scenario → Live → Outcome → Architecture → Pipeline

Lead with the question, show the four sub-agents, then walk the architecture (Fivetran + Iceberg/Glue + dbt Labs + dbt Wizard + Snowflake/4 engines) and the Pipeline page with real Fivetran connector deep-links.

BUSINESS USER · 40% OF BOOTH TIME

Assessors, Appeals, Finance, Public Portal

After the wizard finishes, drop into the dashboard pages. Show that the same governed gold layer powers the executive read. The build-time AI is invisible to them — they see the answer.

Required follow-up materials

- One-pager · Allegheny County Tax · at fivetran-jasonchletsos.github.io/Fivetran-Demo-Repository/one-pager/tax-assessment/
- Live demo URL · fivetran-jasonchletsos.github.io/tax-assessment-databricks-demo/
- 3-minute walkthrough video (this recording)
- Architecture diagram on the Architecture tab of the demo
- Cross-industry credibility · Pediment Bank

Booth-floor reminder: the demo is built to run unattended on the booth screen. If a prospect walks up mid-playback, do not restart. Pick up the narration where the playback is — that is also the point. The build finishes whether someone is watching or not. That is the entire pitch.

The gold layer gets acted on, not just queried

Activations is the fourth native NewCo stage, right after Transformations. The instant `gold.fct_assessment_current` flags a parcel with an assessed-value jump over 20% year over year, no matching row in `gold.fct_appeals` for the current filing year, and the county's appeal-filing deadline inside 30 days, Activations pushes that parcel straight into Granicus GovDelivery as a segmented subscriber — no reverse-ETL vendor, no second copy of the data, no analyst re-keying names into a spreadsheet once a quarter.

TRIGGER 20%+ YoY value shock, no appeal, deadline ≤30 days	DESTINATION GovDelivery Assessment_Shock_Subscribers	PRE-DEADLINE FILING RATE 18% → 52% assessment-shock parcels, per cycle
--	--	--

WHAT YOU SAY	WHAT YOU SHOW
"Assessment shock drives most of our appeal-season call volume. Right now, the moment a parcel's assessed value jumps more than 20% year over year with no appeal on file, that fact sits in <code>gold.fct_assessment_current</code> — but nobody acts on it until an analyst runs Friday's report."	<code>gold.fct_assessment_current</code> flips a row's shock flag to true on screen — assessed value up 24.6%, no appeal on file, deadline in 22 days.
"Watch this instead. The second dbt marks a parcel 'shock, no appeal, deadline in 30 days,' Activations picks up that segment on the next scheduled sync and pushes it straight into Granicus GovDelivery as a subscriber list — value-change percentage, deadline date, and neighborhood already sitting in merge fields."	Architecture page → Activations panel → /activations-live: Segment Definition → Field Mapping → API Push.
"The outreach team's deadline-reminder campaign fires against a list that's minutes old, not weeks old."	/activations-live → Destination Confirmation table; subscriber count ticks up and merge fields populate within the sync window.
"Last cycle, moving this from quarterly-manual to sync-on-build took the pre-deadline filing rate from 18 percent to 52 percent, and cut projected post-certification refund liability by an estimated \$1.8 million."	Before/After panel → Reverse-ETL hop row: Census + Hightouch replaced by NewCo Activations, native.

Modeled outcome: closing the pre-deadline window is estimated to cut projected post-certification refund-and-interest liability by \$1.8M per appeal cycle and pull roughly 3,100 avoidable calls out of the assessor's call-center queue during the 40-day appeal window. Allegheny County Tax is a fictional demo entity — synthetic data only.